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Hot nudges on hazy landscapes

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Abstract
Algorithmic decision-making systems trained on massive data sets, or "assistive AI," has the potential to remove biases and errors in traditional decision-making, according to some behavioral paternalists. Assistive AI's democratization of expert advice represents as important and socially beneficial a technological advancement as the internet's democratization of consensus knowledge. Like any other tool, assistive AI has limitations. I model individuals as theory-based decision-makers whose social systems are open-ended and evolve through time—like "hazy landscapes" with unclear horizons. I consider both traditional nudges and "hot nudges," automated nudges programmed to learn how to effectively influence their targets by collecting personalized data. I demonstrate that in open-ended social systems "hot nudges" on "hazy landscapes" can exacerbate the knowledge deficits of traditional nudges and may have uniquely pernicious effects on advancing non-beneficial policies and suppressing the emergence of beneficial social institutions.

Keywords Nudge theory · Artificial intelligence · Behavioral paternalism · Algorithmic governance · Open-ended systems · Assistive AI

1 Introduction

Nudges as relatively non-invasive policy tools have spread quickly through democratic governments since the publication of Cass Sunstein and Richard Thaler's (2008) book, *Nudge: Improving decisions about health, wealth, and happiness*. Nudges are interventions into the individual decision-making process intended to correct patternable and predictable cognitive biases, errors, and contextual illusions theorized to limit rationality. Behavioral economics and behavioral public policy have gone from

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niche to common academic subfields whose bias-exploiting techniques are broadly employed by both private and public entities.

Economists tend to study nudges intended to "make choosers better off, *as judged by themselves*" (AJBT) (Thaler and Sunstein 2008, p. 5; italics in original; Thaler 2015). AJBT nudges are not easy to design: they should in theory effectively exploit choice contexts and psychological biases to improve the outcomes of the target if and only if the target benefits from the nudge, and should be relatively costless to avoid by individuals who would not benefit from the nudge (Sunstein 2015a). These requirements, inherited from economic theory's fixed-point optimization of individual and social benefit, place formidable epistemological obstacles in the designer's way: the nudge designer must demonstrate individual benefit according to the subjective preferences of the individual and the absence of negative externalities, assuming stationary conditions (Sugden 2016). Furthermore, though this exercise is often neglected, nudges should be shown to be robust or adjustable to radical uncertainty wrought by open-ended, dynamic change.[1]

Nudging must also be justified institutionally, as a sociotechnology intending to make not just individuals but societies better off through the exploitation of bias. I use *sociotechnology* in the sense of Trist (1978), where a sociotechnical system consists of a group of people and a certain way of doing things—a sociotechnology—which impacts individual- and group-level outcomes. Making unguided decisions is one kind of sociotechnology for the production of choices and the social outcomes of choices; being guided paternalistically by policymakers to choices deemed better as judged by the individual themselves is a different sociotechnology for the production of choices and the social outcomes of choices. The feasibility of particular sociotechnologies and their outcomes depend on the institutional framework. Self-debiasing methods might get crowded out by nudges, resulting in inferior outcomes if self-debiasing would have been less costly and more effective as in "small-group debiasing" (Rizzo & Whitman 2019: 213–18). An example of the possible inferiority of individual nudges relative to small-group debiasing in certain contexts is a married couple separately asking their married friends for advice about their conflict instead of attempting to work through it together. While knowledgable about marriage in general, their friends lack both local knowledge about this particular marriage that may be crucial to solving the issue, and the close-yet-different perspective of the spouse.

Presenting nudging as just one of many sociotechnologies for better decision-making departs from the representative agent framing of traditional economic theory. The theorist is required to think in terms of systems and their emergent properties—that the aggregation of many agents isn't simply their sum, but something "more", to paraphrase Philip Anderson's (1972) observation that, in complex systems, "more is different."[2] Sociotechnologies like self-debiasing methods and decision-assisting

[1] The stationary scenario is tractable beneath a certain threshold of system complexity, but scenarios wherein it is necessary for nudges to adapt to dynamically changing conditions are not generally tractable unless they are probabilistically simple, i.e., where the distribution of possibilities and their probabilistic weights are learnable and there is no true novelty or "radical" uncertainty in the Knightian sense.

[2] Such observations go at least as far back to Adam Ferguson's description of social orders without design, where "Every step and every movement of the multitude, even in what are termed enlightened ages, are

institutions are the something "more" that can emerge through the usual channels of evolutionary knowledge and trade to palliate a system rife with biased and suboptimal decision-making.

That nudges can displace or prevent the emergence of decision-assisting methods and institutions is an especially important issue as algorithmic decision-making grants to humanity new tools to assist in complicated-but-comprehensible "noisy" decision scenarios in criminal justice, human resources, healthcare, supply-chain management, even construction (Zhou et al 2023; Kordzadeh and Ghasemaghaei 2022; Belhadi et al 2022; Gupta et al 2022; Saravi et al 2022; Sharma et al 2022; Rudin et al 2020). Generative artificial intelligence (genAI) including large language models (LLMs) has recently made significant technological strides, passing a host of academic exams (de Winter 2023; Katz et al 2024), resulting in a general optimism in the abilities of genAI to assist decision-making in personal and economic settings (Kahneman 2021), governance (Dawson 2024), healthcare (Haupt and Marks 2023), and other fields. In this article I call decision-assisting algorithms "assistive AI," including all kinds of adaptive algorithmic decision-assisting systems, whether they are LLM-based or not.[3]

The optimism of behavioral paternalists about assistive AI isn't deterred by concerns of a potential "algocracy," that is, governance of human affairs by algorithms (Danaher 2016). "What can be done with AI is to create a program that observes an individual's choices. That program will be better than people at a wide variety of things. In particular, it will make better choices for the individual. Why? Because it will be noise free… You should replace humans by algorithms whenever possible" (Kahneman 2018). How do we know when the beneficial algorithmic substitution of some human decision-making process is possible? Kahneman does not elucidate, though it would seem that when and how it is possible to beneficially substitute algorithmic for human decision-making is the very knot to be untangled. Human brains are not like meat computers and do not think solely in the probabilistic, backwards-looking manner of AI and decision-making algorithms. Humans engage in "theory-based causal reasoning" which allows them to connect disparate ideas and project novel ideas from a fantastic future not predicted by data (Felin and Holweg 2024; Felin et al 2024). Nor are algorithms truly "noise free" in any reasonable sense. Algorithmic decision-making can exacerbate exploitative and manipulative patterns in social interactions like transmitting mis-, dis- and mal-information (Park et al 2023), engage in "dark patterns" (Mathur 2021), and intensify the effective influence of powerful—and fallible—individuals and agencies. This presents us with yet another knot to be untangled before we can argue for a beneficial algorithmic nudge.

Supposing we *can* untangle these knots, it is far from clear they won't get tangled up again in the future as one of a variety of factors changes—the context of reasoning, the individual's worldview and goals, the institutional mileu in which decision-making takes place, and so forth. Furthermore, if Devereaux (2025a, b) is correct in modeling open-ended evolutionary social systems as nonergodic, then the John

made with equal blindness to the future; and nations stumble upon establishments, which are indeed the result of human action, but not the execution of any human design." (Ferguson 1782).

[3] I do not extend my analysis in this paper to "strong" AI, as it remains purely speculative.

Smith of today is not the John Smith of tomorrow. What damage is wrought upon John Smith of the future when being nudged as if he was the John Smith from his past—and how might that damage be compounded when algorithmic nudges become more effective at obtaining their programmed target by virtue of their relative customizability and adaptability?

Closed-ended analysis may help make social systems mathematically tractable but it gives social scientists the illusion they can efficaciously intervene by altering system-level variables (like savings rates) to change system states (like overall welfare) which is generally not possible in open-ended systems (Rosas et al. 2024). Social systems are epistemologically open-ended, in that the interaction between individuals and their world is mediated by mathematical tools insufficient to characterize the set of all technological possibilities in a finite time. In open-ended systems, we see to some distance around us fairly clearly, but the landscape further away looks hazy, trees indistinguishable from mountains indistinguishable from men. To strike a reasonable path through this hazy landscape we must update our understanding of the map with almost every step with take.

To be clear, there is little contention as to whether generative AI could be a useful aid in some arenas of human decision-making. Generative AI can help us make sense of the overwhelmingly massive and growing amount of information freely borne of the internet-era. But the simple employment of generative AI to make a nudge more *effective* (through improved customization, for instance) does not automatically render it more *beneficial*. "Whenever possible" in Kahneman's statement cannot only measure whether an algorithmic nudge fulfills it programmed objective. It must also demonstrate benefit, a much murkier problem than proving effectiveness, a problem whose solution forces us to grapple with local knowledge, the institutional mileu of decision-making, emergent sociotechnologies, and the remaking of individuals and their social systems as they evolve nonergodically.

To understand the full interaction between nudge policies and assistive AI, we must conduct an analysis of the evolutionary benefit of competing behavioral algorithms *within* a comparative institutional framework. The combination of assistive AI and nudge policy in the form of publicly crafted self-reinforcing behavioral algorithms, or what I call "hot nudges," select for and against certain system worldlines in open-ended systems in a relatively effortless, automatic and atheoretical way that may be irresistible. An ecology of self-reinforcing behavioral algorithms is daunting enough to analyze in terms of comparative individual and social benefit. Hot nudges, which operate on a policy level, combine the exigencies of democratic political institutions with self-reinforcing behavioral algorithms. Due to their rapid-looping and automatic nature, hot nudges may exacerbate public choice concerns like the crowding out of potentially better options by mandating a single or small number of decision pathways, institutional status-quo bias, ideological bias, and knowledge incompleteness and expert failure.

This paper demonstrates that hot nudges on hazy landscapes can be a dangerous combination, particularly with the ever-lingering ambiguity of determining when it's possible to replace human decision-making with algorithms. The map of the paper is as follows: Sect. 2 is about how and what algorithms know and how they use what they know to make decisions. Section 3 is about how and what behavioral paternalists

know in open-ended systems, with hints for how traditional and algorithmic nudges go awry when systems are allowed to creatively evolve. Section 4 investigates how the theories of behavioral paternalists and the mechanism of algorithms interact given the institutional mix—i.e., embedded in the rivalrous processes of the market or as a part of top-down planning. Section 5 concludes with a suggestion for behavioral paternalists to "hug" rather than "nudge."

2 Algorithmic behavioral paternalism

In this section, I start off by summarizing the state of the art in nudge theory, primarily to nail down what it means for a nudge policy to be "effective" and "beneficial," so as to better engage with Kahneman's advice to replace human decision-making with algorithms "whenever possible," which can be interpreted as when these two conditions are met. Certainly, if an algorithmic nudge is ineffective but beneficial, the cost of administering it will likely outweigh the unrealized benefit; if it effective but non-beneficial then it fails to increase individual and social welfare and should not be implemented; and if it is both ineffective and non-beneficial it fails to increase individual and social welfare by definition and should not be implemented.

I then summarize the state of algorithmic behavioral paternalism in some of its current implementations, focusing on "assistive AI" as the likely candidate protocol for the algorithmic implementation of a nudge.

3 Summary of the state of art in nudge theory

Behavioral theorists in the heuristics and biases tradition have exposed through experiments and empirical studies a large and growing set of ways human decision-making violates neoclassically rational decision-making and therefore fails to maximize individual and overall welfare according to the neoclassical welfare theorems. These violations of rationality are cast by behavioral paternalists as rectifiable faults of decision context, perception, cognition or emotion (Kahneman 2011; Benartzi et al 2017; Congiu and Moscati 2022). The "nudge" theory built atop the theory of heuristics and biases hypothesizes that decision contexts can be redesigned through "choice architecture" to transform a suboptimal choice into a neoclassically optimal choice at negligible cost to the individual. Thus, individuals can be made better off "as judged by themselves" (AJBT) without requiring the theorist to invoke Kaldor-Hicks-style welfare calculations (Thaler and Sunstein 2008; Sunstein 2018). Behavioral biases are analogized to optical illusions, where individuals are said to be making choices in a distorted reality (Kahneman 2011). Nudges are intended to reshape reality into its purportedly "true" and undistorted form, navigating choosers safely and accurately through distorted realities (Sunstein 2018). Additionally, AJBT nudges can be crafted with negligible social cost—they are "nonfiscal"—and with negligible transaction costs, i.e., they are "noncoercive" and preserve freedom of choice (Hertwig & Grüne-Yanoff 2004).

To know when a choice context is distortionary, nudge designers are assumed to be free of bias, to reasonably approximate reality from the perspective of the individual being intervened upon, and to have greater ontological knowledge of the context and system in which choices are made. Nudge designers area assumed to have the ability to determine both how choices are biased and what would have been chosen absent bias. In this way, nudge designers are assumed to be able to guide choosers as would a beneficent parent (or like a GPS).

AJBT nudges are policy tools, emanating from public institutions through legal or regulatory means, and like a parent's guidance are intended to serve as a (non-optional) command. Nudge theorists determine a set of behaviors consistent with optimizing individual utility (under uncertainty) and assign those behaviors to an neoclassically rational agent, nicknamed an "Econ." In contrast, "Humans" are biased and not neoclassically rational. For example, to counter the distortionary context of the bakery, a nudge policy might be to require bakeries to list calorie counts on their menus. To counter the distortionary context of savings decisions, a nudge policy might be to switch employee retirement savings plans from opt-in to opt-out.[4]

Ethical concerns about nudging can be reformulated as methodological or theoretical statements (Sunstein 2015b; Kuyer and Gordijn 2023). A nudge that reduces individual autonomy may do so by removing certain decisions or by increasing the costs of comparing new possibilities to old possibilities (Vugts et al 2020). The ethically neutral standing of an AJBT nudge relies on its "costlessness" to the individual chooser. A nudge is costless if and only if the problem Econs are modeled to solve is the same as the problem Humans actually solve (Sugden 2008). A nudge that circumvents democratic institutions or interacts negatively with political incentives may be a result of unreasonably substituting the subjective preferences of a Human policy-maker for that of an Econ policy-maker, or it may simply be the result of failing to adequately theorize about social institutions.[5]

It's useful to sum up AJBT nudge theory with a few major formal assumptions, particularly if we're going to address Kahneman's advice to substitute human decision-making with algorithms (which would be in this case nudge-type choice context transformations) "whenever possible." Namely,

Assumption 1. Effective AJBT nudges increase social welfare. That is, individuals possess a "true" set of preferences that remain the same throughout their lives; the preference model of rational action a is correct representation of reality; the utilitarian calculus of welfare analytics is a correct representation of reality; a correct representation of reality is (at least, approximately) knowable by nudge designers; nudge designers are without bias.

[4] Nudge theorists rely on context "architecture" to alter the choices Humans make to be more in line with modeled Econs. Nudge policies are openly distortionary, the logic being that choosing like an Econ is welfare-superior and, therefore, exploiting cognitive illusions (like "default bias" in the case of making savings accounts opt-out rather than opt-in) is ethical as long as it's transforming a Human into something more like an Econ.

[5] Particularly their knowledge-storing, choice-coordinating, path-dependent, competitive, dynamic, and emergent characteristics.

Assumption 2. The more personalized the nudge, the more effective the nudge, the better off the nudged, and the fewer negative social costs associated with the nudge.

Assumption 3. The institutional costs of administering correctly designed AJBT nudges through policy mechanisms are minimal, so as to preserve claims that nudge policies increase social welfare. That is, the nudge policy results in an optimal institutional mix or, at least, does not prevent the formation of an optimal institutional mix, with institutions defined loosely, including behavioral heuristics, informal and formal rules, and sociotechnologies to coordinate the provision of public goods like political institutions, firms, mutual aid institutions like fraternal societies, and so forth.

The arguments underlying these assumptions and a rigorous critique is included in Rizzo and Whitman (2019), and won't be entirely rehashed here. I will be referring to these assumptions as needed to contrast the modeled reality of behavioral paternalism, i.e., the reality experienced by Econs, with the reality experienced by choosing agents and intervening experts, i.e., the reality experienced by Humans.

3.1 The tempting effectiveness of assistive AI in complex choice environments

Using massive amounts of highly personalized data to customize nudges is not a new idea. Nudge designers can use massive amounts of data to personalize interventions to make them more effective and beneficial (Sunstein 2013). The amount of data required to make nudges more effective and beneficial increases with the complexity of the choice space, rendering algorithms relatively more effective versus traditional interventions in complex choice environments (Porat and Strahilevitz 2014). In complex choice environments, the amount of personal data required to train effective algorithms would require continuous and invasive surveillance at the individual level, incentivizing the employment of costly-to-avoid technologies like facial and voice recognition to identify aspects of the individual and their behavior to improve the predictability of interventions (Gordon et al 2021). Facial recognition technology, based on deep learning algorithms like convolutional neural networks, is already widespread in policing, border control and airports, for convenience uses like unlocking devices and self-check kiosks at pharmacies and grocery stores, and is being proposed for healthcare settings like assessing pain level, medication compliance, and emotional state (Shakya et al 2016; Jain et al 2016; Phillips et al 2018; Martinez-Martin 2019; Werner et al 2019; Johnson et al 2022; Shore 2022). The accuracy of widespread biometric surveillance like facial recognition has the potential to render public interactions fully non-private, a digital panopticon where deviations from "acceptable" sociopolitical behaviors are easily identified at their smallest scale in a way that circumvents rights to privacy and freedom of expression (Strauß 2023; Palmietto and González 2023; Wang et al 2024a, b).

Assistive AI carries with it the potential to bring this panopticon "indoors" into the home, the doctor's office, work and school. Recall that we defined "assistive AI" as an automated algorithmic decision-assisting protocol that is intended to better realize an individual's particular goals. Assistive AI is a broad category that includes extend-

ing the abilities of disabled and infirm people. My focus in this paper is, rather, on decision-assisting algorithms targeted to average-abilitied people with the assumption imported from behavioral economics that individuals are afflicted with many and varied contextual biases in decision-making. This form of assistive AI is traditionally automated. Assistive AI operates by categorizing large amounts of data using machine learning protocols like clustering algorithms, artificial neural networks, transformer models, and support-vector machines to assist decision-making with minimal expert input (Duan et al 2019). Decision-assistive AI are intended to be prediction machines (Agrawal et al 2022). Custom-solution assistive AI designed to query private corporate databases has been around for some time, but solutions had relatively small and specialized training sets and were brittle to algorithmic failure (Russell and Norvig 2010). Generative AI based on transformer models in the form of chatbot AIs like OpenAI's ChatGPT (GPT stands for "generative pre-training transformer") are trained on massive and diverse data sets, revolutionizing access to assistive AI while resolving many issues with algorithmic failure (Dwivedi et al 2023; Hao et al 2024). While generative AI methods aren't new, the advent of easily available massive data sets and the complex computational infrastructure to derive the initial set of reasonable "weights" between tokens (words) is quite recent. GPT-4 easily passes the bar exam and other professional exams at around 1.8 trillion parameters (de Winter 2023; Katz et al 2024), and speculation on the number of parameters for GPT-5 are in the range of 3.5–17 trillion parameters.

Decision-making can be classified in terms of the complexity of the decision problem. Generally, the more structured the decision, which includes its context if context affects the outcome of the decision-making process, the more amenable decision-making has been to algorithmic automation. Recently, deep-learning techniques, the availability of massive data sets, and a steep increase in computing power have made it possible to use algorithmic decision-making in many unstructured decision problems like in recruitment, parole eligibility, loan eligibility, supply chain mechanics, and healthcare decisions (Duan et al 2019; Belhadi et al 2022; Gupta et al 2022; Puig et al. 2019; Albassam 2023; Haupt & Marks 2023).

Assistive AI with a sufficiently large amount of personalized data on which to train the algorithm and with a well-structured decision problem has the potential to be more effective at helping individuals reach their goals than traditional interventions. Assistive AI can track personal data like social media use, online reading habits, purchases, searches, real-world position, estimated car and transit use, and exercise habits to tailor (micro-target) messages to millions of individuals at once (Matz et al 2024). Thaler and Tucker (2013) believe that making "wise choices" about consumer products is becoming increasingly difficult due to market complexity, and that automated decision-assistants trained on personal data could guide choices away from what experts believe is unwise. Complexity in the choice context is considered bad because it fails to make prominent relevant information that might change behavior. Algorithmic decision-making's recent ability to handled more complex choice contexts make it an obvious candidate to handle complex choice contexts whose very complexity results in suboptimal decision-making.

4 The Human use of human knowledge

Kahneman (2018) calls for algorithmizing all that can be algorithmized. Algorithmizing all that can be algorithmized is logical if errors are "mostly noise," it is possible to identify stable paths of optimal choice sequences, and algorithms are themselves without error or noise (Kahneman et al 2021). The assumptions that make algorithmizing decisions reasonable parallel the assumptions made about knowledge acquisition and use in rational choice theory—that is, if we are Econs, we should algorithmize!

But we are Humans. The usefulness of algorithmic decision assistance depends crucially on how we model target choice contexts, action spaces, and outcomes the target believes are associated with those actions—that is, in how we model the Human use of intractably flawed and incomplete human knowledge. Suppose Overall_Benefit(Outcome A) > Overall_Benefit(Outcome B) for individuals like our target, embedded in some fixed context. When our target executes Behavior X in the fixed context, it leads to Outcome A. Behavior Y, however, is theorized to result in Behavior Y when executed in the fixed context. An algorithm intended to transform Behavior X into Behavior Y by altering the fixed context should result in Outcome B *if and only if* we have properly modeled target knowledge.

Human knowledge acquisition is modeled and algorithmic knowledge is constructed using the same basic logical structure—the binary classification of observations. Epistemological models suppose an individual *i* theorizes about their world by associating observed realities with true/false statements, modeled by static knowledge sets K_i (Kripke 1965; Samet 1990; Aumann 1999a,b) or $K_{i,t}$ for a particular time *t* in dynamic models (Devereaux & Koppl 2024). The logic of these models is simple: they assume individuals engage in a type of theory-based reasoning where reality is composed of objects (physical things, ideas, people, actions, available income) and relationships between objects (A transforms into B, C plus D makes E, driving along route F gets you to Towns G and H). Knowledge is *more* than information in theory-based reasoning, because it must first be understood as a set of true–false statements about any believed-to-be possible state.

Human knowledge models—and large language models—are flawed in that they lack intelligence and explanatory power. In the words of Noam Chomsky, "Their deepest flaw is the absence of the most critical capacity of any intelligence: to say not only what is the case, what was the case and what will be the case—that's description and prediction—but also what is not the case and what could and could not be the case. Those are the ingredients of explanation, the mark of true intelligence" (Chomsky et al 2023). Chomsky et al (2023) provides an example of holding then dropping an apple. Prediction and description are predicting that the apple would fall if one lets go of it, or simply describing what is happening as the action occurs. Explanation, on the other hand, includes counterfactuals like "but for the force of gravity the apple wouldn't have fallen" and causal explanations like "any such object would fall because of the force of gravity" (ibid).

Much of decision theory in economics does not include a dynamic learning protocol. K_i is assumed to be static for all time. This implies that an Econ with incomplete knowledge is doomed to generate suboptimal socioeconomic institutions and of

necessity must be assisted to make better decisions. The theory of asymmetric information is a theory of nonlearning agents whose voluntary exchanges are doomed to be informationally inefficient (Grossman & Stiglitz 1980). Game theory explicitly models knowledge as a semantic process over time as the paradigm allows for knowledge to be constructed and to realize different states, and allows theorists to model situations in which individuals do not know everything they need to make optimal decisions. However, game theory presumes a fully listable and known possibility space for analytical tractability which overspecifies of knowledge (Aumann 1999a,b). Consider the famous Econs Bob and Alice, where to uphold the common knowledge result required to generate predictions in game theoretic models, Alice must know (have in her knowledge set K_i) if Bob knows (has in his knowledge set K_j) whether Alice observes Bob's actions, which is not in general true for Humans—we generally can only know about our own actions, not the knowledge others have of our own actions (Artemov 2022).

In open-ended evolutionary systems, new possibilities are continually added to decision contexts, necessitating a time-dependent individual knowledge set $K_{i,t}$. New possibilities in open-ended systems are irresistible. In open-ended systems we are continually adjacent to real possibilities we do not yet know are realizable, but that we can theorize about— this is the "theory of the adjacent possible" for open-ended systems (Kauffman 1993; Kauffman and Roli 2021; Devereaux, Koppl and Kauffman 2024). Learning may be methodical and hence programmable, but it's often not. Leaps of intuition reveal new aspects of thought-to-be-understood situations; reading an article, poem or book can make weak connections between thought-to-be-unconnected possibilities and change one's entire perspective. Adjacent possibles are dependent on the individual *i* and form a temporal sequence $\mathbf{A}_i = \{A_{i,t}, A_{i,t+1}, A_{i,t+2}, \ldots, A_{t+k}, \ldots\}$ (Devereaux & Koppl 2024). $A_{i,t}$ represents the possibilities realizable in the future at time $t+1$ that we do not yet *know* are realizable at time *t*. So, while the rest of the sequence $\{A_{i,t+1}, A_{i,t+2}, \ldots, A_{t+k}, \ldots\}$ is also made up of realizable possibilities, individual *i* neither knows at time *t* that these possibilities exist nor if they are realizable, and thus cannot theorize about them. The full set of all realizable possibilities is unlistable and, therefore, unrankable. Unlistability and unrankability implies that it is impossible to compute the utility of all possibilities, even without invoking arguments from computational complexity.[6]

In open-ended systems individuals theorize about their worlds in a persistent state of theory incompleteness, where there exist truths about reality that individuals cannot theorize about in their causal inferential framework and must revise their theories to understand. All individuals embedded in a system, including theorists of social systems, are subject to theory incompleteness. The state of theory incompleteness

[6] Model the state of a system as a sequence of zeroes and ones, and the evolution of a system as a sequence of binary-encoded states. Open-endedness means that the smallest program that can describe the states of the system is getting longer over time (Kolmogorov complexity). It is impossible to predict future states of an open-ended system from any earlier state. If we could, the program to describe future states would be embedded in the program to describe the current state. But then the longer future state would have be describable using the shorter program of the current state, which is a contradiction. A predictable system can exhibit stable or decreasing complexity, but it cannot grow the complexity of its states (see Corominas Murtas 2016: p 5).

does not resolve over time due to the continual generation of novelty–-i.e., there is no theory convergence. At any given time, knowledge is inherently local to the individual and therefore fragmented across all individuals, what Devereaux and Koppl (2024) call "frame relativity." Through time, knowledge is inherently fragmented *within* an individual, where later theories are path-dependent but not necessarily consistent with previous theories. Some evidence in favor of fragmented knowledge is that individuals seem to generate preferences for the situation at-hand rather than refer to a fully ranked set of preferences (Lichtenstein & Slovic 2006). The result that individuals can hold mutually inconsistent perspectives on their world throughout their lives is the "many minds" theory of nonergodic knowledge (Devereaux 2025a; Koppl et al. 2015). Furthermore, the path individuals and societies take through the adjacent possible—a path wrought from individual choices—is unmappable and sensitively dependent on choice contexts (Devereaux 2024). What looks like noise to one person could be another person's flash of intuition, "serendipity" (Kirzner 1973), an unalgorithmizable process of knowledge discovery resulting in an overall more beneficial state, or the emergence of a new "mind" with a different perspective and goals. In open-ended systems, there is no way to fully know what is lost when algorithmizing a decision instead of using other assistive technologies, as neither our theories nor our evolving societies can explore all possible outcomes (Koppl et al. 2015; Devereaux et al 2024). If a stable optimal-for-all-outcomes choice sequence identifiable by experts does *not* exist, we can't label Humanlike choice behaviors "errors" or "noise".

Neither are LLMs perfect knowledge classifiers, even in the face of theoretically complete data, leading to famous hallucination problems like the inability for chatbots to consistently and correctly answer questions like "how many r's are in the word strawberry" based on their training methodologies alone. According to a very recent working paper by researchers at OpenAI (Kalai et al. 2025), the reason is similar to how humans resolve decision problems when their theory-based reasoning provides no clear solution: because it is worse to do nothing than it is to guess and potentially get the right answer. Take the classic example of an unclassifiable solution space: Buridan's Ass. Buridan's Ass stands exactly halfway between two bales of hay. If it does not move, it starves to death. Which hay bale should it go for, right or left? Extend that example to an Ass in a dark spacious room. The Ass cannot see farther than a few feet away from her, and she does not see any hay in that direction. There may be no hay or any kind of food in the room, at all. Should she stand still and starve to death because she cannot calculate the probability of hay given a candidate direction of movement, cannot calculate her expected utility, and therefore cannot rank directions? Or should she start moving based on plausible reasoning that has no logical connection to the decision problem—say, moving to the right, because leftward movements are "evil" and hay is "good"? Similarly, LLMs must provide some answer: they are penalized by users for saying they do not know the answer. So they make up an answer using plausible reasoning, hoping that, like students who bluff on exam essay questions they didn't prepare for, refusing to answer means you have no hope of providing a satisfactory answer, while bluffing can, sometimes, be successful. Buridan's Ass doesn't need *right* reasoning to possibly find food, and will

certainly starve if it remains paralyzed by the analytical intractability of the decision problem.

4.1 What problems are Humans actually solving?—A decision theory for Humans

If Humans aren't just *not* doing what Econs do, but *cannot* do what Econs do—that is, if Humans are solving an entirely different problem than Econs—then adjudicating whether a nudge is effective or beneficial depends how one theorizes about choice. In behavioral paternalism, the difference between an Econ and a Human is that the former is a (more) perfect computer than the latter—resistant to cognitive biases and contextual illusions, with a more-than-Human power to take into consideration complex contexts and interactions. If a decision problem is not well-defined, then both effectiveness and benefit must be calculated differently than if the decision problem was tractable to an Econ. A Human choice in a decision problem intractable to rational choice theorizing might appear suboptimal relative to an Econ's choice *as if* the decision problem was tractable. A nudge to alter Human choice to be more in line with modeled-Econ choice in that situation is not, in fact, provably beneficial.

We face the same problem when assessing what it means to replace Human choice by algorithmic choice "whenever possible." As noted in Kalai et al. (2025: 8), hallucinations are not inevitable in LLMs when the decision space is finite, where a fixed number of formulaic questions are asked, when mathematical calculations are well-formed, and when the model can respond "I don't know" to any other question. Simple, stable, robustly consistent, well-defined decision problems could be well-handled by algorithmic decision assistance, particularly if humans retain final agency in any decision. The more complex the decision context, the longer the time period in question, the more dynamically turbulent the decision context and target, the more ambiguous the choice space, the less likely even the most individually customizable algorithmic decision assistance will be able to reliably improve an individual's welfare in their own lights. Algorithms deployed outside a rivalrous process will be even less likely to reliably improve an individual's welfare under these conditions.

The helpfulness of decision assistance in traditional or algorithmic forms depends sensitively on not just how people are observed to behave, but on what they know or believe. Choice architecture depends both on modeling the context but also on modeling how an individual's theory of their world depends on the choice context. In behavioral paternalism, individuals are assumed to be able to maximize their individual utilities even in complex worlds with a large number of possible choices, but may find their calculations thrown off by "distortionary" choice contexts that invoke cognitive illusions and biases. Choice architecture depends on altering the choice context not just to remove presumed distortionary characteristics, but to guide choosers to the presumed-to-be optimal choice.

Assume we are not in the most simple and unambiguous of decision contexts, which appear rarely in this world, but a world that is creatively evolving, where knowledge is fundamentally incomplete. In this more Human world, knowledge is fragmented across individuals and within individuals over time, the full set of realizable possibilities is unlistable and unrankable, there is no theory convergence, and

experts—including choice architects—do not have special access to correct theories or sets of realizable possibilities (Devereaux & Koppl 2024).

Fragmented knowledge leads to truly localized knowledge, wherein knowledge use depends on the possibilities an individual believes are plausible. The combinatorial complexity of decision trees with more than a few possibilities puts an upper limit on how many known possibilities over which an individual can compute (as explained in Footnote 1). In complex contexts, individuals need to store knowledge in easy-to-access forms, including within the configuration of a decision problem itself. Putting a chocolate cake front-and-center in a sales display could encode knowledge relevant to the welfare of the chooser, like that the cake is much loved and demanded by clientele, or that it is past its expiration date and needs to be gotten rid of, or that management is pushing an experimental product that hasn't yet proved its market worth, and so on (unlistably). The placement of the cake could encode knowledge about its quality and taste with respect to the average bakery customer, since it is in the best interest of the bakery staff to coordinate their ends (making money) with the customers' ends (satisfaction). Since many bakeries do this, even if the customer has not yet visited this particular bakery, they may have visited others and had positive experiences when purchasing items on prominent display and can deploy that experiential knowledge in this not-so-new new context.

4.2 Nudging to correct static biases

Static biases are biases invoked in certain time-independent decision contexts. Examples from the nudge literature are profligate and include anchor bias, priming and framing effects. Distortionary choice contexts can be can be narrow and particular, like a dieter deciding between an apple or a slice of cake in a bakery that dresses up their cakes in an attractive display at the front of the shop but stacks their apples in a tiny basket behind the counter, and they can be abstract and a function of immutable properties of the choice context, like a twenty year-old deciding between saving money for retirement, which will not yield positive results for several decades, or taking an exciting international vacation with their friends now.

However, contexts can also encode knowledge. Choice contexts do not exist in a vacuum—they constitute a system of choice contexts that, together, evolve through time, and reasoning in this evolutionary system rather than being neoclassically rational is best understood as being ecologically rational (Smith 2003; Todd & Gigerenzer 2012) An abstract example is a multistage game, where what kind of game (which Elinor Ostrom might liken more to *arenas* of choice than choice contexts) precedes another sequentially can affect the overall payoffs, and whether beneficial versus nonbeneficial equilibria are attainable. Similarly, prior experience affects future choice, which means there is an evolutionary context to decision-making (Hertwig et al 2004). Due to the complex and open-ended nature of knowledge in social systems, truth in the form of an optimal choice cannot be known before undergoing a process of verification (James 1896; Devereaux & Koppl 2024).

To model static choice contexts, consider a behavioral function f_i that transforms states S of individual well-being from one to another through choices and actions. The question to consider is how a certain mix of inputs $p_1, p_2, \ldots, p_k$ contributes to

the generation of a particular desired state, S^*. If we think of the inputs p_j as representing the choice context (i.e., the true/false conditions of relevant characteristics of the world), and the behavioral function as decision assistance in the form of a traditional nudge or an algorithm, then the exercise is to craft a behavioral function that transforms a suboptimal state of wellbeing to an optimal state of wellbeing, such that $f_i\left(p_1, p_2, \ldots, p_k\right) = S^*$. Suppose we indicate whether an individual is nudged by indexing the behavioral function with some value of n (for "nudge"). Suppose that for a certain unnudged individual *i*, $f_{i,n=0} : S\prime \rightarrow S^*$ where $U_i\left(S\prime\right) < U_i\left(S^*\right)$ for all other attainable $S\prime$.

How might a nudge alter the behavioral function? Individual *i* is in state $S\prime$ and wants to realize state S^*. To do that via function f_i, they choose and act to obtain the right mix of values of the p_j. Suppose $p_1 = 1$ means "it is true that I have recently ingested glucose." Perhaps it is crucial that $p_1 = 1$ for the optimal S^* to be realized: in contrast to *Nudge* (Thaler & Sunstein 2008), there is evidence that glucose levels correlate with self-control, and that too-low glucose can lower willpower and increasing glucose levels can restore it (Gailliot & Baumeister 2007; Baumeister et al 2024). Suppose behavioral paternalists craft a nudge $n = N$ to move high-sugar foods from front display cases to hidden storage areas so as to encourage consumption of less sugar, the assumption being that the average customer eats too much sugar and would benefit from eating less. This nudge suppresses the value of the glucose-input p_1 to 0 during the execution of the nudged behavioral function, $f_{i,n=N}$. For individual *i*, who needs the glucose to raise their willpower to, say, study well for an upcoming exam, $f_{i,n=N} : S\prime \rightarrow S''$ where $U_i\left(S''\right) < U_i\left(S^*\right)$ for individual *i*, i.e., the outcome of the nudge results in a *less optimal* outcome for the individual in question. A model is only as good as the state of its science—and in a rapidly updating field like human biology and health, can become quickly out-of-date.

There are two lessons in this example: in rapidly and creatively evolving contexts, even the best crafted nudge can fail to predictably benefit its target; and in populations with diverse worldviews, plans and goals, accounting for true AJBT benefit requires considering individuals that deviate from the average target of most traditional policy interventions. For instance, nudge-type interventions to lower electricity bills differentially impact politically progressive households (Costa and Kahn 2013), and nudges to increase savings rates are negatively impactful for people who have already saved a lot (Thunström et al. 2018).

What does this example have to say about algorithmic nudges? Assistive AI is a knowledge disintermediation tool that improves upon the internet's democratization of human knowledge by giving users a way to hone in on the knowledge relevant to their problem and situation. LLMs have lowered the price of certain customized expert advice to almost zero. Assistive AI's democratization of expert advice represents as important and socially beneficial a technological advancement as the internet's democratization of consensus knowledge. Like any other tool, assistive AI has limitations bound up in its theoretical formulation that can't be overcome with more and better data, like its inability to completely infer and the inevitability of spurious correlations and hallucinations. This is to say that AI cannot, on its own, error-correct all human knowledge knowledge and fill in our knowledge gaps. However, AI is more effective at behavioral decision assistance due to a variety of factors, like

its ability to adapt its advice to the target. We will discuss this in greater length in Sect. 3, but suffice to say that assistive AI has the ability to amplify errors and negative unintended consequences by virtue of its relatively superior effectiveness. For example, once upon a time smoking tobacco was marketed as a way of controlling one's weight, for which it was effective. Of course, there are negative unintended consequences for adopting smoking. Exercising our imaginations for a moment, suppose algorithmic decision assistance existed in this era. When one creates a profile on their fitness app of choice, they can select, "Help me start smoking" in addition to other health goals. The app then sends notifications through one's smartwatch or phone reminding the user to "Have a puff or two for your health!" and helps the user located the lowest-cost tobacco shops nearest to them, and connect up with local smoking clubs. The relative effectiveness of algorithmic decision assistance could have caused smoking to spread much more quickly than it did, with all its attendant negative consequences for general health.

4.3 Nudging to correct time-inconsistent preferences

Possibilities we encounter in any decision context are more expansive than their momentary impact on our lives, or as an observation in some data set. A possibility isn't a point in time and space, it is a line that branches manywheres through time. Possibilities connect us to other possibilities, are process-like, and have benefits and detriments which can change through time.

Knowledge in open systems is fragmented across an individual's lifetime, implying that i might at time $t+1$ revise their theory of reality, goals, and wants to be inconsistent with their presumed reality, goals and wants at time t. An Econ whose preferences are time-inconsistent is biased or irrational (cf. Ainslie 2017). A Human whose preferences are time-inconsistent is not necessarily biased or irrational. Illustrated simply, consider a nudge $n = N$ intended to benefit an individual i in i's own estimation. Suppose i's utility in an unnudged state $S_{n=0}$ at time t is $U_{i,t}(S_{0,t})$, which is less than $U_{i,t}(S_{N,t})$ given that i should be using knowledge like an Econ. A successful nudge is when $U_{i,t}\left(S_{N,t}\right) > U_{i,t}(S_{0,t})$ for all t.

Determining whether a nudge meant for an Econ is successful for a Human is a thorny matter, which we uncover when breaking down the meaning of the above inequality. The unnudged individual is undergoing a process whereby the intertemporal choice context is impactful to their ultimate state at time $t=k$. The sequence of unnudged states is $\mathbf{S}_{i,n=0} = \{S_{0,t}, S_{0,t+1}, \ldots, S_{0,t+k}\}$. Their knowledge set changes over time to reflect interaction with the adjacent possibles at each time t and with theory revision, represented by the sequence $\mathbf{K}_i = \{K_{i,t}(S_{0,t}), K_{i,t+1}(S_{0,t+1}), \ldots, K_{t+k}(S_{0,t+k})\}$. States $S_{0,t+m}$ where the $m>0$ are generated by a process $F(\mathbf{K}_i, \mathbf{S}_i)$ that is a (not necessarily continuous or well-defined) function of knowledge and previous states. Suppose the maximally beneficial state of the unnudged individual is realized in state S^* at time $t=k$, *whether or not* the individual knows this at time t. Suppose the unnudged individual will reach state S^* at time $t=k$, where $F\left(K_{t+k}\left(S_{0,t+k}\right), S_{0,t+k}\right)) = S^*$. A successful nudge of a Human requires that for $F\left(K_{t+k}\left(S_{N,t+k}\right), S_{N,t+k}\right)) = S\prime$, the nudged utility $U_{i,t}\left(S\prime\right) > U_{i,t}(S^*)$. Knowledge impacts more than the problem situation, so it is crucially important that nudged knowledge $K_{t+k}\left(S_{N,t+k}\right)$ is at

least no worse than unnudged knowledge $K_{t+k}\left(S_{0,t+k}\right)$ *in all its other and future revisions.* The recursive complexity of this problem cannot be understated, and it is discussed at great lengths in works critical of nudge theory's central claims (like Rizzo and Whitman 2019). Unfortunately, nudge theorists rarely address the effect of nudge on learning processes and knowledge whatsoever, much less definitely demonstrate that *given* how nudges alter an individual's knowledge and state over time, the altered processes *will result* in an improved state of existence for an individual in their own lights.

Furthermore—and cripplingly so—once we fragment an individual across their lifetimes, it does not make sense to compare their intra-lifetime utilities just as it doesn't make sense to compare the subjective preferences of one individual to another. We cannot expect intertemporal consistency of a person's worldview in an open-ended system–-the best we can hope for is local consistency during a short timeframe and coherence between adjacent states, but the complexity of the possibilities known to an individual make even this an idealized situation (Devereaux and Koppl 2024).[7]

Algorithmic interventions, due to their superior relative effectiveness, could lock individuals into temporally suboptimal decisions more quickly and for longer. Even as it becomes evident that the advice is diverging from changing individual preferences, the adaptability of assistive AI "learns" how to reign in behaviors supporting diverging preferences and reverting them to supporting old preferences. I tackle this issue directly in the next section.

5 Hot nudges on hazy landscapes: algorithmic governance through decision assistance

A behavioral policy may be effective without being beneficial. Failure modes for algorithmic decision assistance can be generally classified into three categories: model failure, expert failure, and algorithmic failure. These failures can be modeled as the "noise" in algorithmic decision assistance, contra Kahneman's claim that algorithms lack noise (and by virtue of their noiselessness stand to be better decision-makers than humans). Model, expert, and algorithmic failure modes affect or are affected by the institutions informing their methods, rules, and implementations. In this section I cover these three failure modes in an attempt to address Kahneman's implicit question of when it is possible to substitute human decision-making with an algorithm, his explict assertion of the relative noiselessness of algorithms, and how the institutional mix supports the welfare content of algorithmic decision assistance more generally.

Steering individual decisions and their presumed social outcomes ultimately boils down to picking the winners—or, more accurately, picking how to pick the win-

[7] I don't include a discussion of "boosts" in contrast to nudges (Hertwig & Grüne-Yanoff 2017), as the analysis goes through in a similar way: rather than assume the decision context riddled with noise and errors, boosters see the individual's theory itself as riddled with noise and errors, fixable by some intervention at the level of knowledge. Boosts privilege one way of understanding the world rather than a certain set of preferences, as in traditional nudge.

ners. The primary debate in economic policy circles is whether relying on decentralized rivalrous processes or conducting explicit engineering exercises results in the best individual and social outcomes. Rivalrous processes are governed by no single person, but require a certain ecology of rules that protect private property, enforce contracts and allow anyone to initiate consensual trading relationships (Smith 2003). Explicit engineering requires a high centralization of power and relatively weak private property rights, so as to allow for the easy disposal of resources by powerful social engineers.

Under rivalrous processes, the ultimate winners are uncertain, but the outcome is rich with the knowledge of an entire society, as directly and indirectly taking this knowledge into account is the only way for winners to emerge (Kirzner 1997). An outcome rich in knowledge is an outcome that must of necessity have taken into account a great many of the perspectives, wants and goals of the individuals in the society in which it emerged (Lavoie 1985). Cellphones evolved from bulky car-bound boxes accessible to the top 1% of income-earners, to delicate screens through which an individual can access the sum total of human knowledge for around 1% of the average monthly income.[8] Engineered solutions—solutions developed or selected through political processes—tend to freeze in place out-of-date or just plain wrong decisions. Ready examples are the adoption and persistence of Lysenkoist methods of agriculture by the Soviet Union, and later, "Great Leap Forward" Maoism (Borinskaya et al 2019); central standardization of Soviet computing practices in the 1960s-70 s leading to a lag behind Western computing advancement; and the widening difference in patenting activity between present-day East and West Germany, implicating the legacy of socialist institutions in East Germany's history of technological development (Fritsch et al 2023).

Human decision-makers don't always get it right. They aren't always trying to learn or self-debias to improve their decision-making, and this can have negative effects on social welfare more extensive than whatever consequences individuals suffer for bad or willfully ignorant decision-making. Human institutions have evolved in part to manage when the consequences of bad decision-making affects others, and much ink as been spilled over what kinds of insitutions manage this task better. Designating a particular institution to managing the task has its own consequences, depending on the nature of the institution and its relationship with alternative institutions for managing the bad consequences of decision-making.

For instance, consider the "biased" choice to not save "enough" for retirement when one is young. If we suppose, like nudge theorists, that this is a widespread bias exhibited by enough people to warrant a nudge at the policy level, then the knowledge of this quirk of humanity will be encoded in the institutional framework in a variety of formal and informal responses. Mutual aid networks are one response, emanating from fraternal, professional, and religious institutions (Beito 2000). Most of these institutions aren't explicitly engineered, their forms and practices emerging incrementally and within a competitive ecology of insitutional options over time.

[8] In 2025, the monthly cost of a single line internet plan is between $40-$70 (low-end is prepaid, high-end is post-paid) and the average monthly salary of an individual in the United States as of 2025 is around $5,200 (US Bureau of Labor Statistics 2025).

One response requires coercive public institutions that mandate saving, but whether that solves the problem requires observation and public choice reasoning. From a public choice perspective, mandated saving could entice public officials to "borrow" from that big new pool of cash (as they have, from Social Security), it could suppress the development of a superior sociotechnology for encouraging retirement saving through crowding out and path dependence, and it could place an undue burden on individuals who would benefit from saving less as younger people, even if it's impossible to know—even from an individual's own point of view—who these individuals are.

Similarly, algorithmic decision assistance comes in different forms and may have different implications for individuals and societies depending on whether it is voluntarily adopted as one of several rival algorithms competing in a market versus whether it is politically constructed/adopted then imposed on a market.

5.1 Effective, but not beneficial: the mismatch between theory and practice in algorithmic behavioral policy design on hazy landscapes

In creatively evolving systems, path dependence is a problem in both rivalrous and centralized winner-picking processes, and no one possesses an oracle that reveals the best or better paths a society can take.

Each path represents a different alternate future that moves us towards some goals and away from others. In creatively evolving systems, the socioeconomy as we know it, paths "become" into existence in the Bergsonian sense, a process of self-creation that can never be written down. Open-ended systems feature "hazy landscapes" with finite prediction horizons beyond which neither expert nor algorithm can see. The metaphor of hazy landscapes invokes neoclassical control theory, where the goal is to move an economy from some lower to higher beneficial "peak" represented by identifying the full set of stable attractor states (cf. Acemoglu's *Modern Growth*).

In hazy landscapes, a policy modeler cannot see how fate twists beyond the prediction horizon, they see only the exigencies of the moment, and they may unwittingly obliterate a worthier policy path as a result. As a society, we may never know we picked a loser instead of a winner. If the landscape is "hazy"—we cannot see what would have happened absent the intervention—failures might not even be detectable, as there is no modelable counterfactual. AI is heralded as a way of improving human decision-making at all levels, including informing better policy by means of being informed by more extensive data. But data is a plural noun in the past-tense, in that even at its best, it is always backwards-looking. In open-ended systems there is no way to design for all contingencies, because not all contingencies can be known to anyone in the system (Taylor 2019; Banzhaf et al 2016; Corominas-Murtra et al. 2018; Devereaux et al. 2024). Like all policy design, it is not about choosing the best path through a known map but about choosing one path out of a few paths that can be illuminated with our historical and theoretical flashlights and ignoring all others. We see but a small part of the map around us and through time. And distortions are everywhere, borne of path dependence, dark patterns and other exploitative behaviors, incompleteness and inconsistency, the sheer costliness of choice, cognitive limitations, fashion and whim. We will discuss a number of these distortions presently.

5.2 Expert failure and bias as "noise" in algorithmic decision-making

Assistive AI, currently in the form of chatbot LLMs, has had a profound impact on the field of expert advice by significantly lowering its price—sometimes to zero, depending on the field of inquiry. Expert advice can be about what goals to achieve or the steps to achieve a meta-goal (like improved overall health), or how to realize those goals. Expert failure is all around us and is unavoidable under conditions of incomplete scientific knowledge (Koppl 2018) and in conditions where choice contexts creatively evolve (Devereaux et al 2024). In the absence of a complete vision of the landscape, even beneficent policymakers must by necessity substitute their own vision for both the ontological truth and for the epistemological perspective of the target individual.

Consider a non-algorithmic behavioral policy, food labeling. Food labeling is generally effective in nudging consumption away from what the Food & Drug Administration (FDA) doesn't want citizens to consume—sugar, salt and fat—and reduces caloric consumption overall (Shangguan et al 2019). This assumes that sugar, salt and fat are universally bad for health and that people universally eat too much. The scientific consensus is not at all unified on these questions, having changed its positions drastically over the years, from backing high-carbohydrate low-fat diets in the late twentieth century to relatively low-carbohydrate and higher-fat diets in the twenty-first century (Bazzano et al 2014). Animal fats were implicated in the mid-twentieth century in heart disease and partially hydrogenated oils (PHOs) were believed to be "healthier," while our current scientific belief is the opposite, that trans-fatty naturally present in animal fats are not harmful and trans-fatty acids in PHOs so harmful that PHOs have been largely banned (Pipoyan et al 2021). Neither is dietary salt as pernicious as once thought—a small fraction of non-hypertensive people are "salt sensitive" with blood pressure increasing with excessive salt intake, and about 60% of hypertensive people are salt-sensitive, calling into question the need for a population-wide nudge away from salt (Haddy 2006). For the groups who need to consume more calories for health, a small reduction in calorie consumption is more harmful in terms of mortality risk than it is beneficial to those who should arguably consume less (Li et al 2021).

The feedback cycle for public dietary advice and regulation is fairly slow, though not as slow as the legislative process. Meanwhile, the market has responded to the demand for off-label dietary advice as the science has changed—like advice promoting a low-carbohydrate diet. Diet-focused apps and software have made it possible to specify desired daily grams of carbohydrate, fat, and protein, and track food consumption in terms of those metrics through the apps. The versatility of the apps in that they do not adhere to any given standard but can suggest a variety of candidate standards gives users a tool to help make more beneficial dietary decisions in light of new evidence, and, importantly, to customize those suggestions with respect to individual dietary restrictions (like for lactose-intolerant individuals). Rivalrous processes seem to operate fairly well in this domain of decision-making.

However, consider our example of smoking for weight control in Sect. 3. Suppose this advice is dispensed in nonrivalrous fashion, and enforced in the same manner through policy-level behavioral nudges to smoke more. The institutional mix in this

case promotes rent-seeking by tobacco companies, but also regulatory capture by these wealthy companies, giving them the political power to suppress or de-influence new science suggesting that smoking has deleterious consequences. While the history of the rivalrous tobacco industry is rife with rent-seeking and capture, the relative effectiveness of algorithmic decision-assistance combined with the rents that go along with politically enforced behavioral interventions stand to do much more damage than a traditional intervention or traditional rent-seeking from companies subject to rivalrous processes.

5.3 Algorithmic failure: hallucinations, dark patterns, crowding-out, deskilling, and algorithmic bias

Algorithmic learning shares modal logic and probabilistic reasoning with our model of Human knowledge acquisition (see Russell and Norvig 2010: Chapters 12–15). Algorithmic learning updates an existing corpus of knowledge using a representation of reality that associates events with actions, typically in a probabilistic manner, and optionally with an error feedback protocol (ibid: Chapters 18–19). Supervised learning is when a modeler provides a schematic training set that instructs how the algorithm should understand inputted data—like pictures of cats and dogs labeled by the modeler to train the algorithm to recognize cats and dogs on its own. Unsupervised learning is when a modeler provides an algorithmic method for how the data should be understood instead of direct examples—like by noticing events that tend to happen at the same time or in the same geographical location, or by noticing words that are more likely to occur together, or some mix of the two. Reinforcement learning integrates reward/punishment mechanisms into its feedback protocol. Large language models are semi-supervised in that they use cluster analysis on word sequences in a pre-training stage, then are subjected to "supervised fine-tuning" on a set of curated prompt-output pairs to instruct the LLM how they should be responding to prompts. Reinforcement learning is used in conjunction with human feedback to further improve the performance of the LLM (Huang et al 2023a, b).

Algorithms are neither noiseless nor error-free. Generative AI frequently "hallucinates," where hallucinations represent "generated content that is either nonsensical or unfaithful to the provided source content" (Huang et al 2023a, b). We can only know what is nonsensical or unfaithful to the provided source content if we possess the correct answers and relevant sources and know how sources are being used. In any large enough data set there are a very large number of spurious correlations (Calude & Longo 2017). Deep-learning neural networks like used in generative AI rely on spurious correlations as part of their primary protocol, and are fragile to situations not predicted by them. Image recognition AI recognizes cows by correlating them with their pictured environment. If they are trained on images of cows in grasslands, they will fail to recognize cows standing on a beach (Geirhos et al 2020). The mitigation of spurious correlations in AI depends on fully mapping the possibility space (Zheng et al 2024), which is implausible in combinatorially complex spaces and ambiguous datasets, and impossible any kind of open-ended system evolution—that is, AI faces a classic knowledge problem. AI cannot on its own distinguish between what

is spurious and what is plausible, and since neither can science distinguish between all possible spurious and plausible correlations, neither can any expert-trained AI.

Just as whether a traditional nudge effects its intended goal is not evidence the goal is beneficial, whether an algorithmic nudge effects its intended goal better than traditional nudge or voluntary exchange is not evidence the goal is beneficial. The effectiveness of assistive AI may be evidence of assisting an individual reach a goal that made them better off in their own lights, but it is also possible that assistive AI is merely a good manipulator whose algorithms are harder to ignore than traditional salience nudges like food labeling. Dark patterns are assistive AI intended to nudge users away from making a choice not in the best interest of the algorithm designer, used notoriously to make users click through screens answering questions or wait on hold to cancel an online subscription it took a single click to sign up for (Ehsan & Riedl 2024; Mathur et al 2021). Personalized nudges powered by AI can help people choose stronger passwords (Peer et al 2019) but can also help email scammers seem more believable (Weidinger 2022) and dark patterns, darker (Park et al 2023). AI deception outscales human deception in sheer breadth and rapidity.

In discussing the state of the art of AI deceptive capabilities, Park et al (2023) defines deception as "the systematic production of false beliefs in others as a means to accomplish some outcome other than the truth." Deception requires both the production of false beliefs in others and the intent to realize an outcome that is unrelated to a truthful realization by those others. Playing devil's advocate to engender a realization in favor of truth isn't deception in this definition. Nor would AJBT nudges that exploit biases be considered deception, as they are intended to realize the same outcomes for biased individuals as would have been realized for individuals who are fully "rational." However, an undeceptive intent in programming an algorithmic AJBT nudge can become deceptive in its implementation in the case of model, expert, and algorithmic failure. In an incomplete knowledge environment, nudging towards some truth means favoring some beliefs over others. Dark patterns exploit the same psychological biases (negatively) that nudge-type interventions purport to exploit (positively). One individual's genuinely helpful assistive AI is another individual's dark pattern.

Algorithmic failure is not just pernicious, but catching. AI bias has been shown to "infect" humans who use it, with humans independently reproducing an AI's bias after they used the tool (Vicente & Matute 2023). This demonstrates that assistive AI doesn't just provide additional knowledge, but substitutes for—crowds out—other ways of acquiring knowledge. Generative AI can produce plausible-looking fraudulent articles publishable in top medical journals (Májovský et al 2023). Algorithmic behavioral paternalism may ironically lock societies into a state without robust knowledge generation institutions or debiasing methods, continually dependent on algorithms for decision-making. "Deskilling" is a known result of implementing automated decision-making algorithms in leiu of human decision-making. Decision deskilling is when decision assistance that is too broad, intense or frequent may remove crucial incentives for learning and growth that improve the overall decision quality of the chooser. No choice architect wishes to doom her target to bias by means of subverting bias.

Algorithmic bias is another known failure mode of automated decision-making, where the opacity and complexity of algorithmic decision-making leads to unintended biases in resultant decisions (Kordzadeh et al. 2022). An example of is in healthcare, where an algorithm intended to predict additional support needs in patient populations systemically underrepresented the additional support needs of Black patients due to confusing health care costs with access and need (Ratwani et al 2024). Algorithmic bias represents a kind of "noise" in the system that is not possible to fully overcome, as the predictions of algorithms are only as unbiased as their training data (Davenport et al. 2020; Sun et al 2020). Algorithmic bias also emanates from biases in algorithmic design, development and deployment ("method bias") and from persistent social and historical biases present in training data ("societal bias") (Akter et al 2021). From a knowledge perspective, error and ignorance cannot be defeated by algorithms trained on human data.

5.4 The peril of over-automation: "hot" nudges and the role of human feedback

Algorithms are tools with the ability to target interventions, hypercustomize to individuals and their circumstances, learn how to better intervene at the individual level, and act and improve rapidly and automatically. AJBT nudges are intended to be beneficial to those they can help and relatively costless to avoid by those they cannot help. Unfortunately, much of what makes algorithms more effective at producing certain behavioral outcomes makes them costlier to avoid, and therefore more pernicious when the justification for the intervention diverges from individual preferences or from scientific truth. The tools of algorithmic governance have given policymakers new ways of ensuring compliance with the plan represented by a policy path, and assistive AI, new ways for behavioral paternalists to ensure compliance with the behaviors determined in their models to realize some higher welfare "peak" of the landscape.

However, the relative effectiveness of algorithmic governance and assistive AI in ensuring compliance contains the seeds of its prospective failure. Algorithmic governance can be more effective than some forms of traditional governance due to the ways in which it is unlike traditional governance: in its ability to mass- and micro-target interventions and automatically revise rules of engagement to address noncompliance and circumnavigatory adaptations. For better or worse, the rules of how policies are implemented are largely not subject to public comment beyond basic legal and constitutional protections, as they are handled by bureaucracies. Bureaucratic slowness is much decried and mocked—in a popular animated movie from the 2010s, bureaucrats were depicted as literal sloths[9]–but its slowness in improving compliance rates makes unintended consequences relatively easier to avoid. Given that it is a general feature of democratic systems that a policy is easier to adopt than to reform or repeal, it is better if a policy with significant unintended negative consequences is relatively ineffective and thus easier to avoid. And while bureaucracies have the power to handle issues like implementation and compliance by altering delivery methods, they do not have the power to alter the intended goal of the rule, even if negative

[9] I refer to the sloths who worked at the registry of motor vehicles in the movie *Zootopia* (2016).

unintended consequences arise that place the beneficial nature of the goal—which in economic theory can be how goals are selected in the first place—into question.

In open-ended systems, unintended negative consequences are a perpetual concern as knowledge is perpetually incomplete. Consider what I call "hot policy mechanisms" implemented to both algorithmically evaluate whether individuals are adhering to some set of rules and to craft customized interventions for those individuals. These hot policy mechanisms get even hotter when they can evaluate the success of their intervention and modify their methods when unsuccessful. The idea of a hot policy mechanism is close to that of a cybernetic system (Wiener 1960), except that it recursively implements a goal at the individual level rather than operating at the level of an entire system, i.e., at the level of goal-making. Goal-making is instead extrapolated to policymakers. Rather than reviving cybernetics, as is argued by Yeung (2017), behavioral paternalists want to craft effective mechanisms for realizing goals whose beneficial nature is determined by some other means, either by classical arguments in economic theory, a consensus of experts, or the democratic process.

Hot policy mechanisms are generally not repairable before the accumulation of significant negative consequences—hot policy mechanisms are fragile to error. Hot policy mechanisms can affect a large number of people and vital social functions in an almost instantaneous fashion, and require a relatively small number of people to be constructed and deployed. Algorithmic governance allows an increasingly smaller number of people to effectively influence an increasingly larger number of people. The danger of rogue branches of government using algorithmic nudging to deploy effective propaganda or alter citizen behavior to be more in line with political agendas via Big Brother-style surveillance cannot be understated.

I've identified four issues with "hot nudges":

Issue 1. Epistemological thinness of centralized, cybernetic designs compared to social innovations made in a thick and fast-moving knowledge ecology
Issue 2. Negative feedbacks with a politicized institutional environment
Issue 3. Pernicious effects of algorithmic failure under algorithmic governance
Issue 4. Unintended consequences of selecting against welfare-increasing resistance against bad policy, like civil disobedience

Algorithmic governance substitutes not just for traditional governance, but for social entrepreneurship and self-organized solutions to social problems. Effectively influential algorithmic behavioral paternalism could lead to the substitution of the subjective valuations of target populations for the normative values of nudge designers (Issue 1). Furthermore, it could lead to the destruction or crowding out of beneficial social institutions (Issue 2). Effectively influential algorithmic nudge could alter citizen beliefs and behavior to be more in line with political agendas (Issue 3), where behavioral deviations like civil disobedience are themselves prevented by effectively influential algorithmic nudges (Issue 4).

Algorithmic behavioral paternalism tends to exacerbate the problems with behavioral paternalism in open-ended systems—those "ethical" problems that are really methodological and theoretical. Its problem is precisely in its relative effectiveness. Take for instance an existing purported consensus that overweight and obesity are bad

for health and can be prevented (see e.g. Tremblay et al 2010). Behavioral paternalists often cite overweight and obesity and sedentary lifestyles in particular as obvious cases that both prove bias and the need for an external debiaser. "Unhealthy" eating is one of the foundational examples of hyperbolic discounting used in behavioral economics (Epstein et al 1991; Loewenstein 1996; O'Donoghue and Rabin 1999; Thaler and Sunstein 2008).

Algorithmic behavioral nudges find some of their earliest commercial realizations in fitness apps, where users are asked to commit to a fitness goal, are provided recommendations based on their existing health data and are served reminders to exercise, drink water, and record daily caloric intake. These apps are cited for having a lack of adequate behavioral support (read: nudges) and as such do not result in much additional weight loss over traditional methods (Puig et al 2019). The useful metric in digital interventions is "engagement," whose units are in attention that has a positive opportunity cost (Nahum-Shani et al 2022). An engaged user is expending effort in—paying attention to—messages delivered by the behavioral algorithm rather than using that time to engage with something else. Engagement is the currency of the attention economy, where much of our interactions with algorithms takes place. Apps with more behavioral nudging—like a higher frequency of messages nagging users to weigh in or record intake—are more effective in helping users lose weight (Gilmartin & Murphy 2015; Valle et al 2020).

Epistemological thinness is a concern in algorithmic governance is (Issue 1), so let's dig into the supposed consensus that overweight and obesity are bad for health and can be prevented. A major comorbidity of obesity is hyperinsulinemia, which can result in chronic diseases like Type II diabetes. The common wisdom, shared by much of the medical establishment, is that behavior causes overweight and obesity (Taubes 2013) which causes hyperinsulinemia. In fact, the causality is likely reversed, with hyperinsulinemia causing overweight and obesity (Erion and Corkey 2017; Templeman et al 2017). Neither is the link between behavior and obesity ironclad—about 75% of obesity is explained by genetics (Elks et al 2012), with behavior and epigenetics accounting for the rest. Overeating can cause weight gain, but only temporarily. Dieting using any method can cause weight loss but most if not all of the weight lost is regained within five years (Nordmo et al 2020; Fothergill et al 2016; Tobias et al 2015). The etiology of obesity is more complex than commonly believed, with two major camps in medical research being those who believe obesity is primarily a result of excessive energy intake (Hall et al 2022) and those who believe it is primarily caused by "fuel partitioning" that is "trapping" fat into storage rather than deploying it for active expenditure. The latter theory is strongly linked with the presence of risk alleles for the fat mass and obesity-associated (FTO) cluster of genes (Huang et al. 2023a, b). Fuel partitioning in obese people who have the risk alleles for the FTO cluster—what I'll call FTO-obesity hereafter—results in a tendency to store fat rather than to burn it due to an imbalance in types of fat tissues favoring storage-type "white" fat over expenditure-type thermogenic "brown" fat.

What is the harm in behavioral interventions when they are not warranted? Weight-cycling, when weight is lost and then regained, can have many negative consequences for metabolic and mental health (Thillainadesan et al 2022; Madigan et al 2018; Wang et al 2024a, b). As intentional weight loss is largely temporary, yet

obese individuals continue to be expected to lose weight by most family doctors, obese individuals are likely to weight cycle over their lives. This situation could be particularly risky for FTO-obese individuals, with each cycle shifting body composition of fat more and more into the relatively more dangerous white adipose tissue. Effective behavioral interventions to lose weight in the short term, made more effective by being algorithmically delivered through apps, could exacerbate the severity of each and frequency of cycles as initial weight loss becomes "easier." Rather than making obese people non-obese and healthier, an effectively influential algorithmic behavioral intervention built on the consensus view could make many obese people less healthy.

NudgeRank is an example of a health-nudging app employed at the public level that has been deployed to over 1 million individuals via Singapore's public Health Promotion Board in a kind of mass experiment (Chiam et al 2024). NudgeRank an algorithmic protocol "designed to foster positive health behaviors on a population scale" (ibid). What constitute positive health behaviors are determined by the programmers of the NudgeRank algorithm and are not customized by the individual, despite the app's characteristic-level customizability based on pre-existing health conditions and demographic factors. The developers report algorithmic success in terms of how well the algorithm helps users to realize pre-programmed goals. For our FTO-obese individual, NudgeRank may help them more effectively lose weight in the short term, but it cannot prevent inevitable metabolic reactions like rebound cycling or the negative health consequences of weight cycling. And while FTO-obesity could be eventually programmed as a pre-existing condition with its own customized recommendations when consensus knowledge catches up, the goals of algorithmic behavioral interventions can only move as fast as the scientific consensus.

Assistive AI is currently present in other areas of healthcare decision-making, though typically more on the provider-sider. Algorithms trained on patient data have been used by Medicare insurers to predict length of hospitalization, with benefits cut off on the predicted date regardless of whether the patient is determined by the doctor as needing more time in hospital. Individuals have already suffered from being unwittingly subjected to error in algorithmic cutoffs, being released while still needing in-hospital care (Narayanan and Kapoor 2024: 9). Decision support systems in healthcare provide recommendations for patient care to a clinician during an office visit and recommendations to surgeons and diagnosticians (Sharma et al 2022). Assistive AI in its most beneficial form be designed with flaws inherited from incomplete knowledge, but an additional concern is that assistive AI can be programmed to favor the interests of insurance companies, hospital administrations and bureaucrats over patients and their doctors.

6 Concluding thoughts: On "hugging" instead of nudging in an algorithmized landscape

We might conclude that this paper doesn't rule out substituting algorithms for human decision-making in Kahneman's view—as "whenever possible" rules out situations of algorithmic failure or bias. But Kahneman's qualification, when taking into con-

sideration the failure modes of AI, is so broad that it becomes meaningless. It is like saying that we should let an expert decide for someone only when expert advice provably benefits the target. The entire gnarly problem, of course, is in proving it. Similarly, Kahneman's assertion of substituting algorithms for human decision-making "whenever possible" moves the target to the gnarly problem of proving when algorithmic decision-making benefits the target, given its failure modes. That is, his assertion gives us little direction whatsoever.

What we do know from our analysis is that the relative agility of algorithms is a strength in bottom-up applications due to the competitive pressures of rivalrous processes, but is a *weakness* in public policy implementations. The agility of algorithmic governance will tend to amplify errors without the corrective pressures of public comment or competition. A financial example is the 2010 trading algorithm "flash crash" that wiped out $1 trillion of market value (Borch 2016). The automatic nature of algorithms could crowd out the emergence of genuine social institutions or alternative decision-support technologies or the development of individual knowledge to adapt to problematic decision contexts. Living under a system of intrusive monitoring and coercion transforms "every act of kindness [to] an act of personal score maximization, and is perceived as such by the recipient… Or worse, the very concept of a voluntary act of kindness gradually becomes just a fading memory of something people used to do. Visiting an ailing friend in hospital will, under such a system, have no more moral significance and emotional value than stopping at a red light" (Russell 2019: 106). Facial recognition AI are already more accurate than humans at detecting IQ, emotions, political orientation (Kosinski 2021; Kosinski et al. 2024), and sexual orientation (Wang & Kosinski 2018). Creating effectively influential behavioral tools to intervene on behalf of the individual can turn into tools for social stratification, psychological manipulation, and propaganda dissemination at the turn of a political administration.

The most pernicious kinds of oppressions are enshrined in law. An analysis of data on defendants in Broward County, Florida has revealed that automated decision-making in parole decisions has resulted in a racial bias against black parolees, classifying them as higher risk (Angwin 2016). Facial recognition technology in China has been used to suppress protests, has resulted in a spike of arrests for various crimes against people, and is being used to exclude people on the massive social credit blacklist being maintained by the Chinese Supreme Court and which seems primarily targeted at dissidents and tax evaders (Devereaux and Peng 2018).

Furthermore, the relative effectiveness of algorithmic interventions may lock society into these suboptimal states as deviations are effectively "corrected" and civil disobedience, stopped before it can begin. Even presuming assistive AI has beneficial goals, assistive AI are only as accurate as the data in their training sets. Data collected relevant to social systems tends to be heavily biased towards the status quo (Boracas & Selbst 2016). Individual adaptation around pernicious rules, where we define the adaptation as welfare-increasing, will become more difficult. Given a behavioral problem to solve, artificially intelligent planners may remove, prohibit or otherwise excise the behavior rather than solve the problem (Cohen et al 2024). If algorithmic governance is so effective as to lock society into a set of suboptimal states, the social system will lose its ability to beneficially order itself.

Biased news aggregation algorithms operating on major social media sites are concerning for good reason, but it is relatively easy to avoid social media sites. Facial recognition cameras on the corner of every city block and in every subway station and train car, the forced installation and use of certain apps or the banning of alternate apps (like how China funnels citizens into the politically controlled and censored WeChat app), and the legalized exfiltration of private data cannot be easily avoided. Glimpsing the future and potential ways algorithmic governance can be employed, algorithms to track spending and bank information administered by a public authority cannot be easily avoided, nor can algorithms administered by a regulatory body to surveil home energy use and restrict it at certain hours, nor can algorithms administered by bureaucracies to surveil of the purchasing behavior of benefit recipients that is restricted if "improper" use is discovered. We can uninstall a fitness app for exfiltrating our geographic data to authorities; we cannot uninstall algorithms that exhibit negative unintended consequences for us (or intended dark pattern behavior) if they are administered by a public entity who has made them legally mandatory.

Algorithmic decision-making processes may be productively employed in governance, but they must be employed in such a way as to make them subject to resilient, innovative, and adaptive forces in human-co-created social systems, to reinforce the epistemological processes of a social system, not to inhibit, destroy, or replace them. Instead of nudge, I propose a "hug." My definition of "hugging" is the *embrace* of advancement. Encouraging advancement is a peer-to-peer relationship rather than a teacher-student, expert-nonexpert, or parent–child relationship, putting interventionists on equal ground with those-to-be-intervened-upon. If it is true that noise and error and bias afflict everyday decision-makers, then it is also true that noise and error and bias afflict paternalists. The doers and done-upon should acknowledge that they stand on equal epistemological ground and seek to advise or be advised, but never to manipulate or be manipulated. The path not taken may be the path someone needs to take, and as experts we must reserve the freedom to ourselves and to those targeted by our policies to opt out of them.

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